



ME 300: Engineering Thermodynamics

Example: Isentropic Steam Turbine

Example

Determine the flow rate required to produce 25 MW of shaft power from an ideal (adiabatic, reversible) steam turbine in which the steam enters at 10 MPa and 720 K and exits at 5 kPa.

Find: $\dot{m}$

Given: $\dot{W} = 25 \text{ MW}$

$P_1 = 10 \text{ MPa}$
 $T_1 = 720 \text{ K}$

h_1, s_1

$P_2 = 5 \text{ kPa}, s_2 = s_1$

h_2

Assume: $\Delta s_{ke} = \Delta s_{pe} = 0$

ad + rev $\rightarrow$ isentropic $\Rightarrow s_1 = s_2$

steady-flow

S.C.S.

$$\Rightarrow \dot{m} (\cancel{\Delta h} + \cancel{\Delta s_{ke}} + \cancel{\Delta s_{pe}}) = \cancel{Q} - \dot{W}$$

$$\dot{W} = -\dot{m} \Delta h$$

$$\dot{m} = -\frac{\dot{W}}{\Delta h}$$

↑ find h_1
 h_2



What assumptions are appropriate for this problem?

- A. Steady flow
- B. $\Delta ke = \Delta pe = 0$
- C. Adiabatic
- D. All of the above

Example

State 1 : $P_1 = 10 \text{ MPa}$
 $T_1 = 720 \text{ K}$

D.2 $\rightarrow T_{\text{sat}} = 584.15 \text{ K}, T_1 > T_{\text{sat}} \Rightarrow \text{vapor}$

D.3P $\rightarrow h_1 = 3233.7 \text{ kJ/kg}$
 $s_1 = 6.4098 \text{ kJ/kg}$

Example

State 2 : $P_2 = \underline{5 \text{ kPa}}$

$$S_2 = S_1 = 6.4098 \text{ kJ/kg}$$

$$\begin{array}{l} D.2 \rightarrow S_f = 0.4716 \text{ kJ/kg} \\ S_g = 8.4012 \text{ kJ/kg} \end{array} \left. \vphantom{\begin{array}{l} D.2 \rightarrow S_f = 0.4716 \text{ kJ/kg} \\ S_g = 8.4012 \text{ kJ/kg} \end{array}} \right\} \begin{array}{l} S_f < S_2 < S_g \\ \underline{\text{mixture}} \end{array}$$

$$X_2 = \frac{S_2 - S_f}{S_g - S_f} = \frac{6.4098 - 0.4716}{8.4012 - 0.4716} = 0.749$$

$$h_2 = X_2 h_g + (1 - X_2) h_f = (0.749)(2560.15) + (1 - 0.749)(136.42)$$

$h_2 = 1957.8 \text{ kJ/kg}$

Example

$$\dot{m} = \frac{-\dot{W}}{h_2 - h_1}$$

$$= \frac{-(25 \times 10^6)}{(1951.8 - 3233.7) \times 10^3}$$

$$\Rightarrow \boxed{\dot{m} = 19.5 \text{ kg/s}}$$



Would a non-ideal turbine produce greater than, less than, or the same amount of work as an isentropic turbine?

- A. Greater than
- B. Less than
- C. Same